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mm-Wave Acoustics: Tackling a Fundamental Limitation of Acoustic Wave Technologies

Milad Koohi

04/04/2025

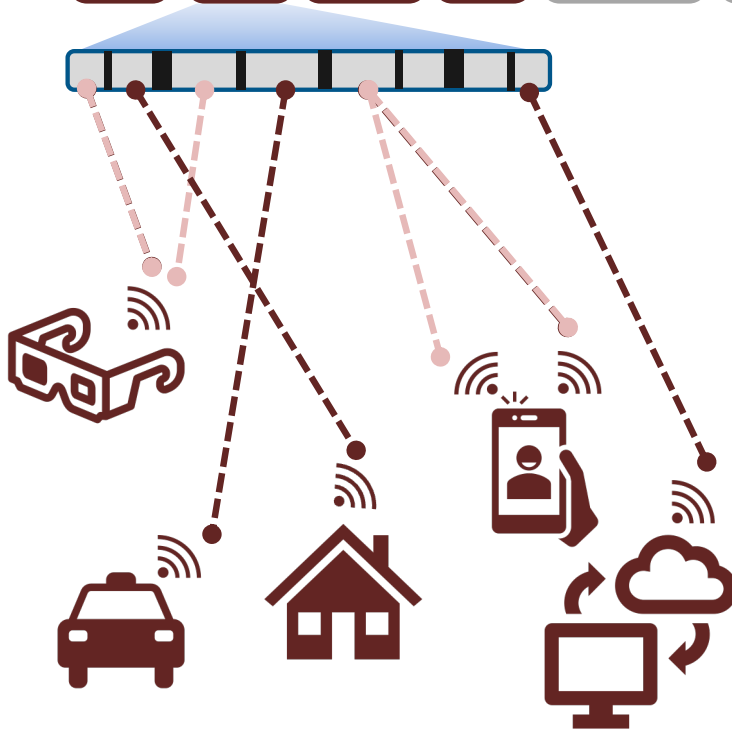
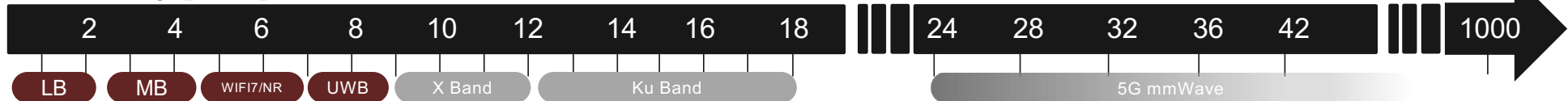
Broad Agenda



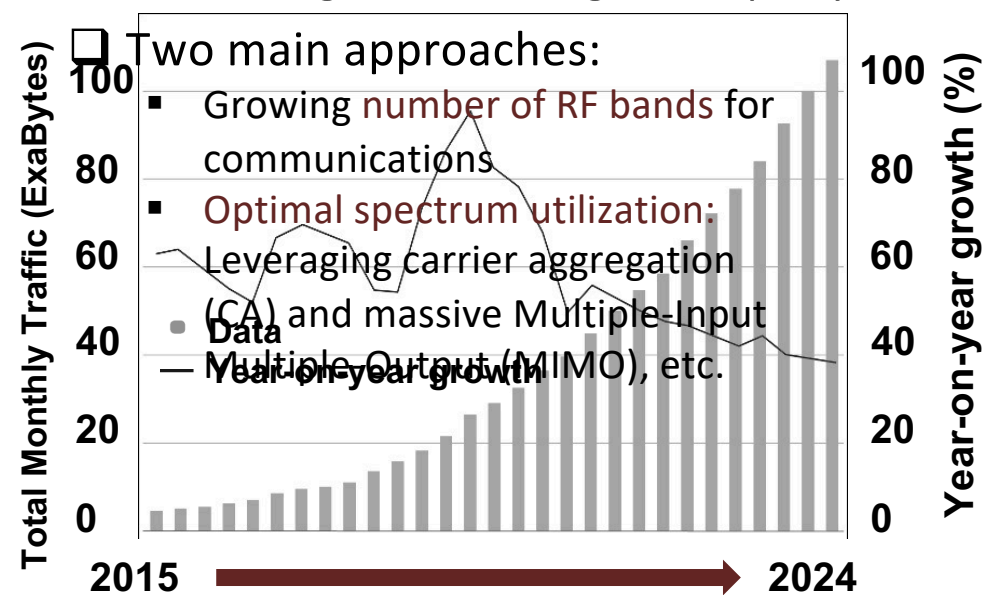
- Challenges and Motivation
- Why mm-Wave Acoustic is Challenging?
- How to break the fundamental frequency limit to enable mm-Wave Acoustics?
- Conclusion

Challenges and Vision for Communications and Sensing

Frequency [GHz]

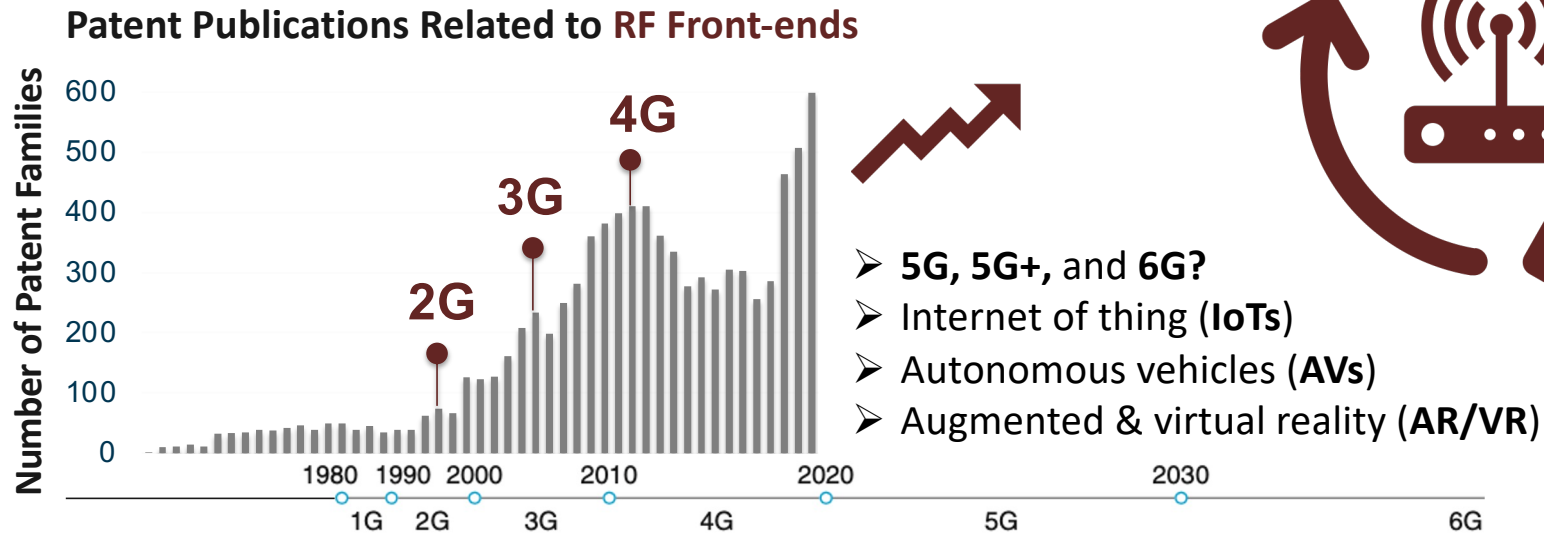


On average 50% data growth per year!



[Ericsson Report 2024]

Challenges and Vision for Communications and Sensing

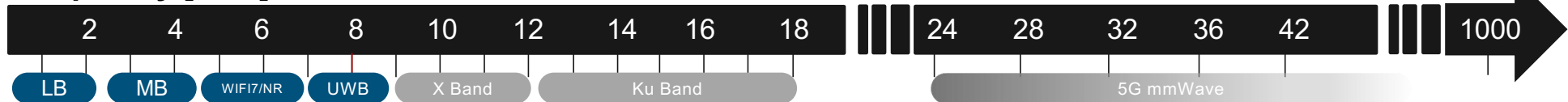


Challenges and Vision for Communications and Sensing



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Frequency [GHz]

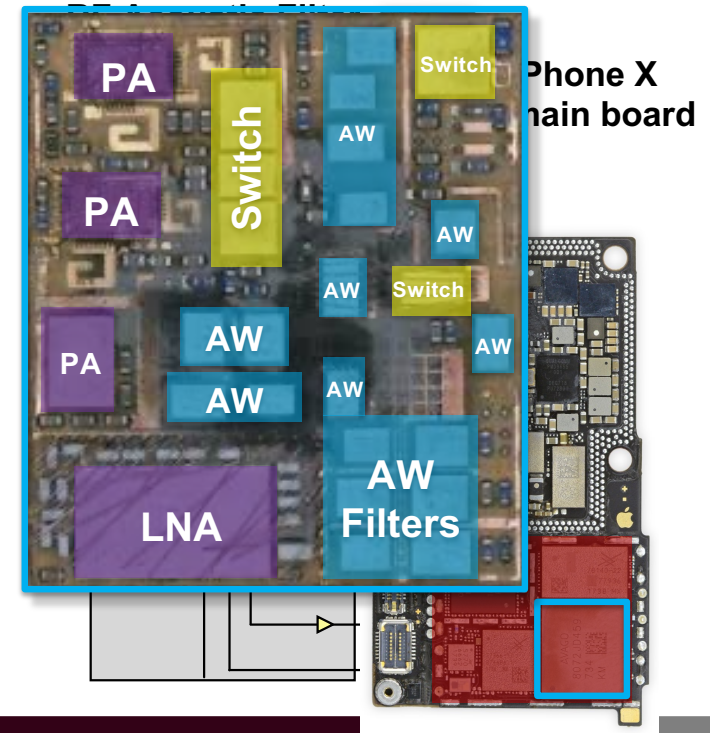
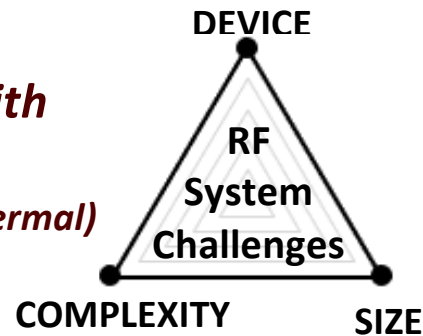


Current RFAW Technologies

Challenges:

- ❑ Growing number of components in RF modules is increasing system complexity and size
- ❑ Device limitations: For example, the frequency of the current RF acoustic wave (AW) device technologies is limited to <8 GHz

Multifunctional materials with coupled domains
(Electromagnetic, Acoustic, Optical, Thermal)

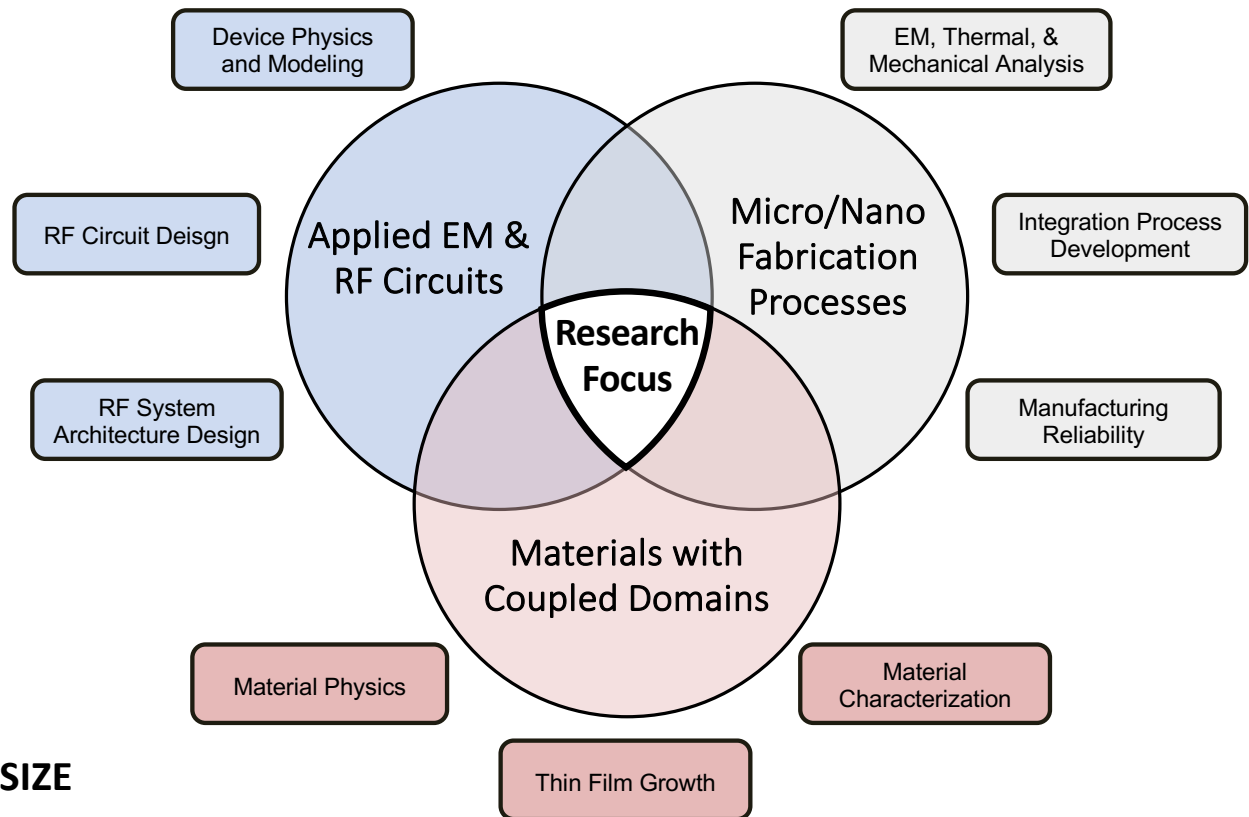
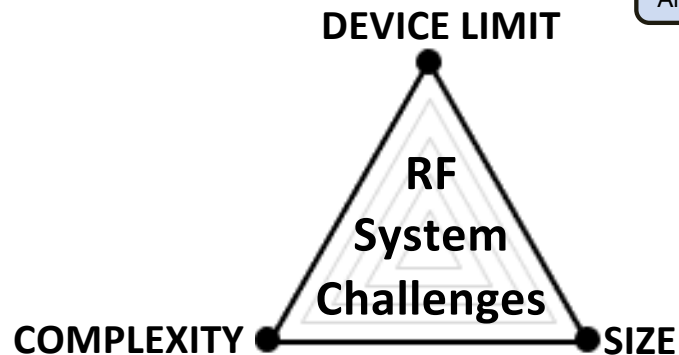


Challenges and Vision for Communications and Sensing



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Vision: To develop enabling technologies for the next generation communication and sensor systems by exploiting multifunctional materials



Koohi Research Group

RF Technologies: from Materials to Systems!



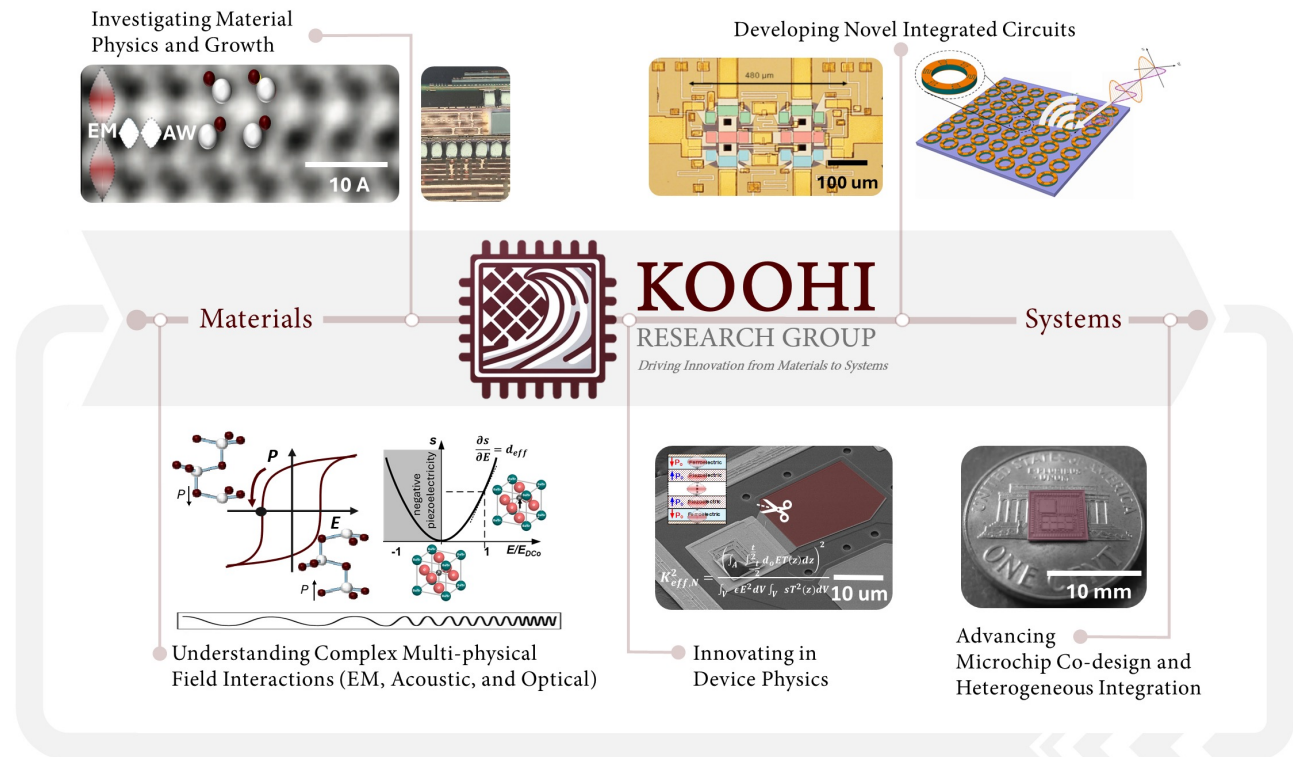
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Research Interests:

- RF Electronics & Systems (MHz – THz)
- Multiphysical Microelectronics & Microsystems
- Advanced Materials: Piezoelectric/Ferroelectric III-N & Complex Oxides

Applications:

- Communications: 5G-advanced and 6G
- Space communication, navigation, and synchronization





Why mm-Wave Acoustic Integrated Circuits (AWICs)?

Why AWIC?



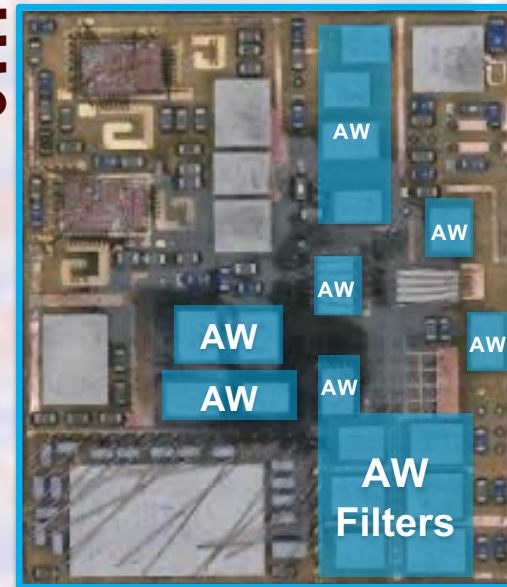
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Frequency [GHz]



Why coupled electromagnetic and acoustic domains?

- ☐ Applications of RF AW integrated circuits: filters, oscillators, gyroscopes, sensors, quantum, etc.
- ☐ Why EM waves are being converted to AWs?
 - ✓ **AWs are 5 orders of magnitude slower than EM**
→ e.g., 1 GHz wavelength: 3 μm (AW) vs 30 cm (EM)
 - ✓ **Extremely low loss at GHz frequencies**



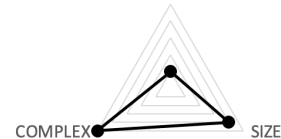
iPhone X
main board

Sub-6 GHz AWIC Example



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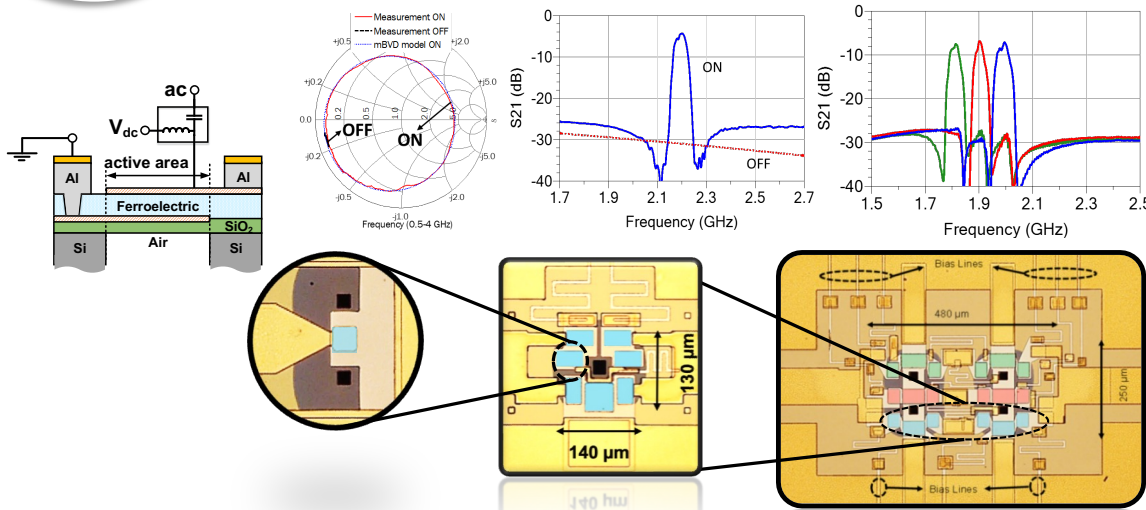
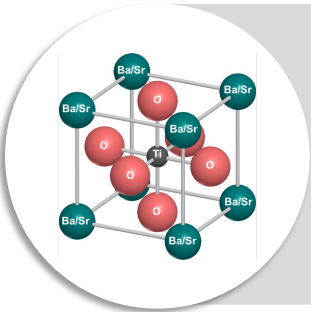
DEVICE LIMIT



RQ: How to reduce the complexity of the filter module?

Selected Contributions:

- ☐ Intrinsically switchable AW resonators and filters with record figures of merit
- ☐ The first demonstration of frequency and BW reconfigurable acoustic wave filter modules

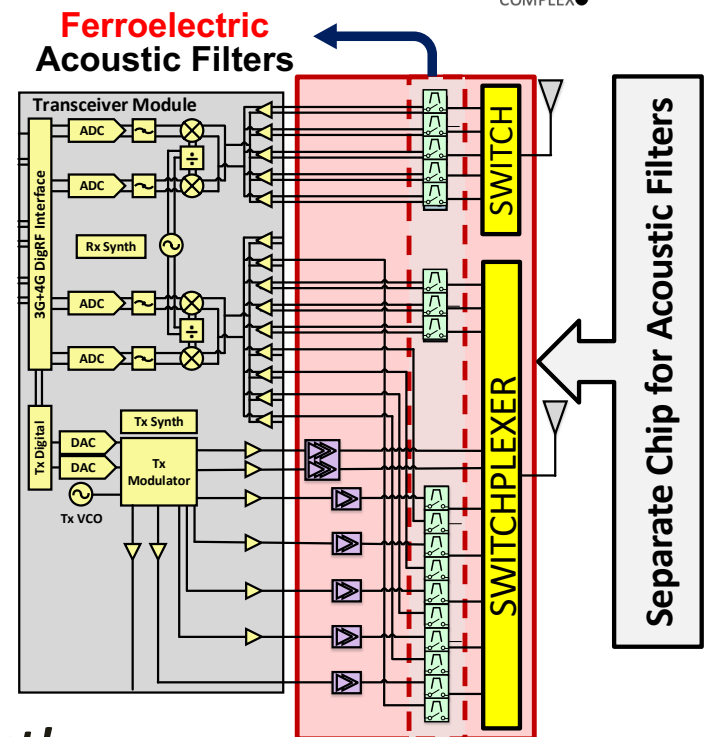


[M. Koohi, et.al. *IMS* 2017]

[M. Koohi, et.al. *TUFFC* 2017]

[M. Koohi, et.al. *TMTT* 2018]

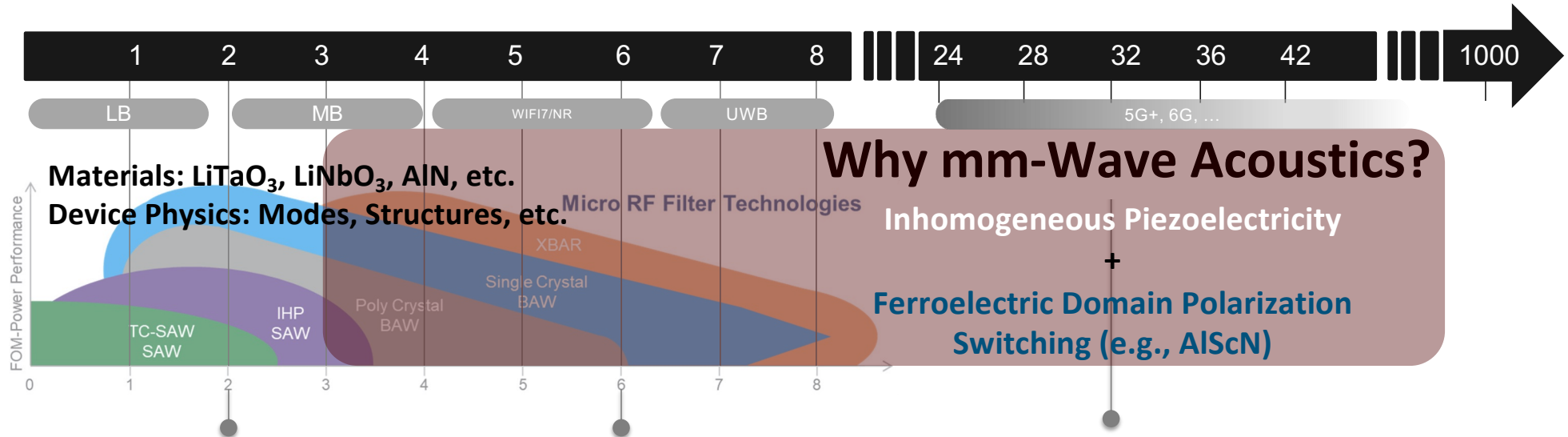
Electrostriction instead of traditional piezoelectric effect!



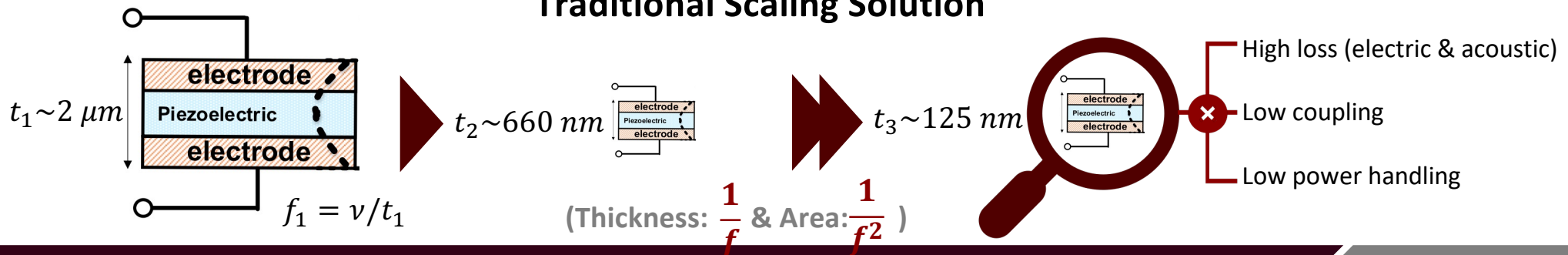
Mm-Wave Acoustic Device Challenges



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Traditional Scaling Solution



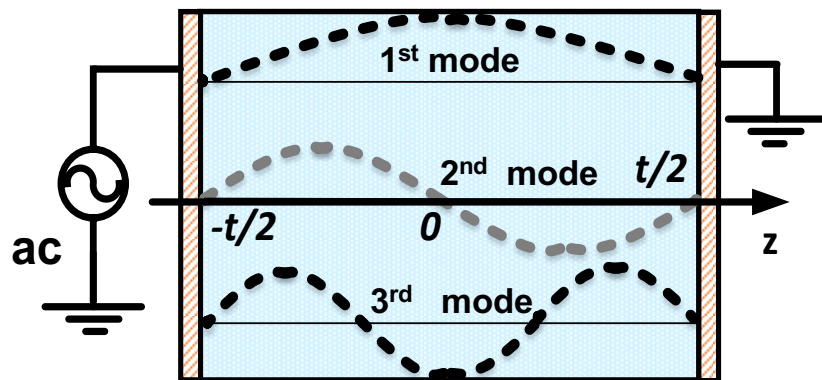
Harmonic Resonance Modes (HRM)?



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❑ **RQ:** Can we use harmonic resonance modes?

“Harmonic Resonance Modes”



Fundamental Limitations:

- ❖ Coupling to AW harmonic resonance modes drops with $1/f^2$

$$K_{eff}^2 \propto 1/f^2$$

Fundamental Limitation of HRMs



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❖ Where is this fundamental limitation coming from?

$$T = cS - eE$$

$$D = eS + \epsilon E$$

$$S = \frac{\partial u}{\partial z}$$

$$\frac{\partial^2 u}{\partial z^2} + \frac{\epsilon \rho}{(c\epsilon + e^2)} \omega^2 u = 0, \quad \frac{\partial T}{\partial z} \Big|_{z=\pm \frac{t}{2}} = 0$$

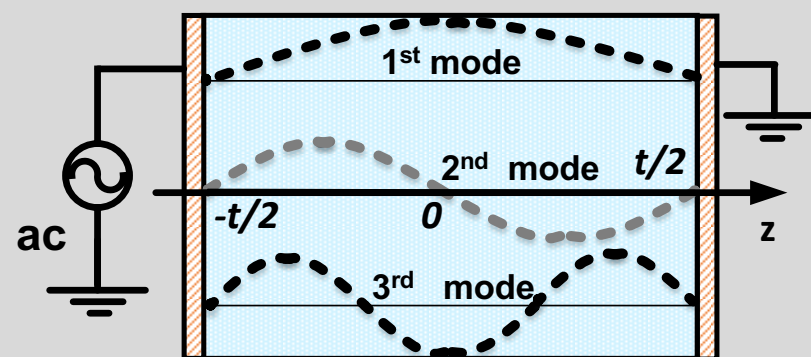
$$u_n(z) = u_o \tilde{u}_n(z) = \begin{cases} u_o \sin(k_n z), & n = 1, 3, 5, \dots \\ u_o \cos(k_n z), & n = 2, 4, 6, \dots \end{cases}$$

$$k_n \frac{t}{2} = n \cdot \frac{\pi}{2}$$

Berlincourt Electro-mechanical Coupling Definition:

$$K_{eff}^2 = \frac{U_m^2}{U_e U_d}$$

$$U_m = \frac{1}{2} \int_V d_o (TE) dV, \quad U_e = \frac{1}{2} \int_V s T^2 dV, \quad U_d = \frac{1}{2} \int_V \epsilon E^2 dV$$



$$K_{eff}^2 = \frac{\left(\int_A \int_{-\frac{t}{2}}^{\frac{t}{2}} d_o E T(z) dz \right)^2}{\int_V \epsilon E^2 dV \int_V s T^2(z) dV}$$

$$= \frac{1}{2s\epsilon \left(\frac{t}{2}\right)^2} \left(\int_{-\frac{t}{2}}^{\frac{t}{2}} d_o \tilde{u}'_n(z) dz \right)^2$$

[M. Z. Koohi, et. al., TUFFC 2020]

Fundamental Limitation of HRMs



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❖ Where is this fundamental limitation coming from?

$$K_{eff,n}^2 = \frac{1}{2s\epsilon \left(\frac{t}{2}\right)^2} \left(\int_{-\frac{t}{2}}^{\frac{t}{2}} d_o \tilde{u}'_n(z) dz \right)^2$$

Conventional homogenous
piezoelectric analysis

$$\begin{aligned} K_{eff,n}^2 &= \frac{d_o}{2s\epsilon \left(\frac{t}{2}\right)^2} \left(\int_{-\frac{t}{2}}^{\frac{t}{2}} \tilde{u}'_n(z) dz \right)^2 \\ &= \frac{1}{n^2} \frac{8d_o^2}{\pi^2 s\epsilon} \end{aligned}$$

Proposed idea: inhomogeneous
piezoelectricity!

$$\begin{aligned} K_{eff,n}^2 &= \frac{1}{2s\epsilon \left(\frac{t}{2}\right)^2} \left(\int_{-\frac{t}{2}}^{\frac{t}{2}} d_o(z) \tilde{u}'_n(z) dz \right)^2 \\ &= \frac{d_o^2}{2s\epsilon} \end{aligned}$$



[M. Z. Koohi, et. al., *TUFFC* 2020]

How to Tackle the Fundamental Limitation of HRMS?

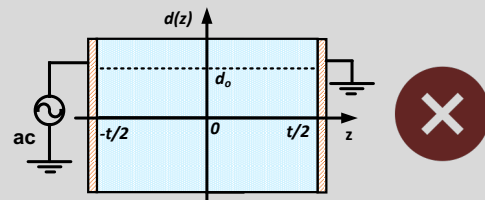
❑ **Fundamental limitation:** Coupling to AW harmonic resonance modes drops with $1/\text{freq}^2$



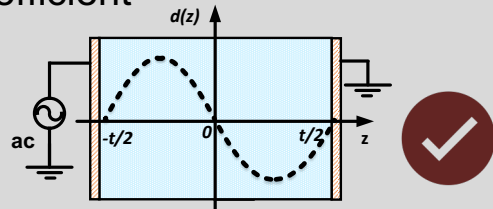
$$\dots \rightarrow K_{eff}^2 = \frac{\left(\int_A \int_{-t/2}^{t/2} d_o E T(z) dz \right)^2}{\int_V \epsilon E^2 dV \int_V s T^2(z) dV}$$

$$= \frac{1}{2s\epsilon \left(\frac{t}{2}\right)^2} \left(\int_{-t/2}^{t/2} d_o \tilde{u}'_n(z) dz \right)^2$$

Fundamental
Analysis



Conventional assumption:
homogenous piezoelectric
coefficient



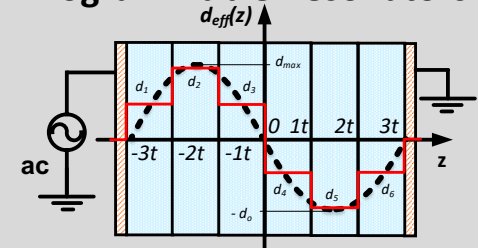
Example of inhomogeneous
piezoelectric coefficient proposed
by my analysis

[M. Koohi, et.al. TUFFC 2020]

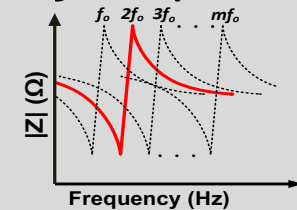


How to realize?

Programmable Resonators:



- multilayer ferroelectric heterostructures
- **Electric-field-induced positive and 'negative' piezoelectricity**



[A. Mortazawi, M. Koohi, US patent]

How to Create Arbitrary Inhomogeneous Piezoelectric Medium?



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1. Electrostriction Phenomenon

2. Ferroelectric Effect

Arbitrary Inhomogeneous Piezoelectric Medium: 1- Electrostriction



Electrostriction (DC E-Field Induced Piezoelectricity)

$$S = \alpha P^2 \rightarrow P = P_s + \chi E$$

$$S = \alpha P_s^2 + (2\alpha P_s \chi + \alpha \chi^2 E) E$$

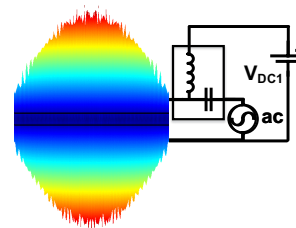
Paraelectric Phase: $P_s = 0$

$$S = \alpha \chi^2 E^2$$

$$\frac{dS}{dE} \bigg|_{E_{DC}} = 2\alpha \chi^2 E_{DC}$$

Acoustic Response

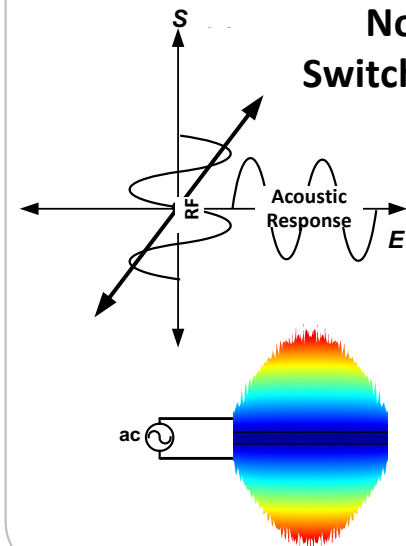
No acoustic response!



Piezoelectricity

$$S = d^T \times E$$

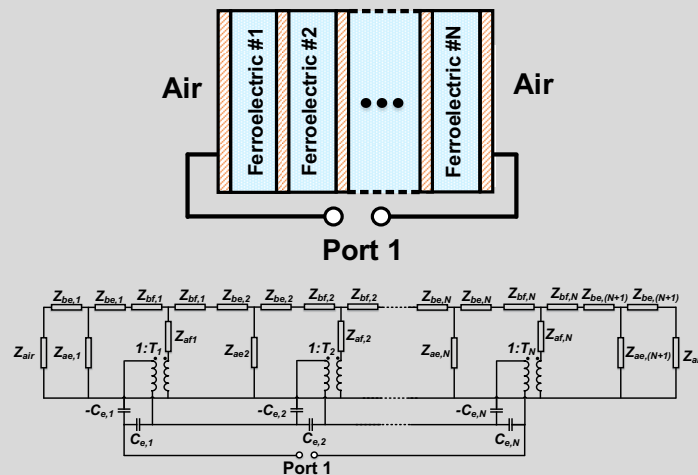
No Switching!



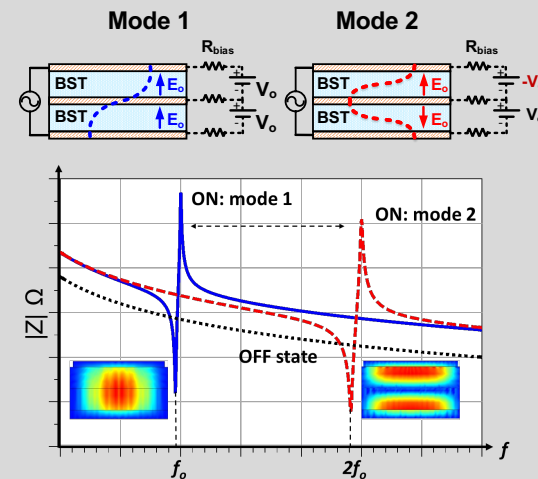
[(Invited) M. Koohi, et. al., *Microwave Magazine*, 2020]

How to Design Programmable Resonators?

- 1D AW lumped element model was developed for multilayer structures (coupled AW & EM domains) :



- Three programmable states:
 - Mode 1 $V/V_{ON} = [1 \ 1]$
 - Mode 2 $V/V_{ON} = [1 \ -1]$
 - OFF state $V/V_{ON} = [0 \ 0]$
- When either of modes is selected the other mode is entirely suppressed

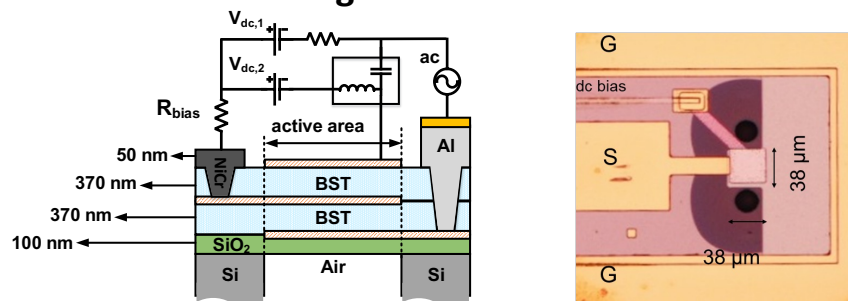


[M. Z. Koohi, et. al., *TUFFC* 2020]

Experimental Data Programmable Resonators

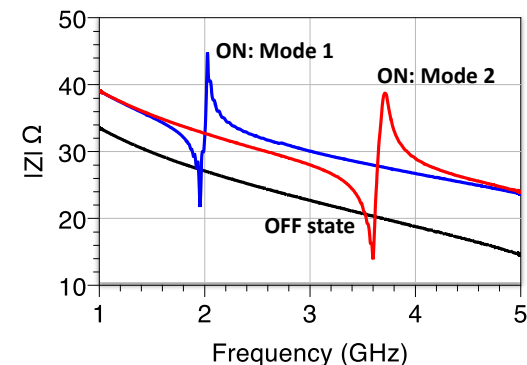
- ❑ **Fundamental limitation:** Coupling to AW harmonic resonance modes drops with $1/\text{freq}^2$

Cross-sectional View and Photograph of a Fabricated Programmable Resonator



[M. Koohi, et. al., *IMS 2019*]

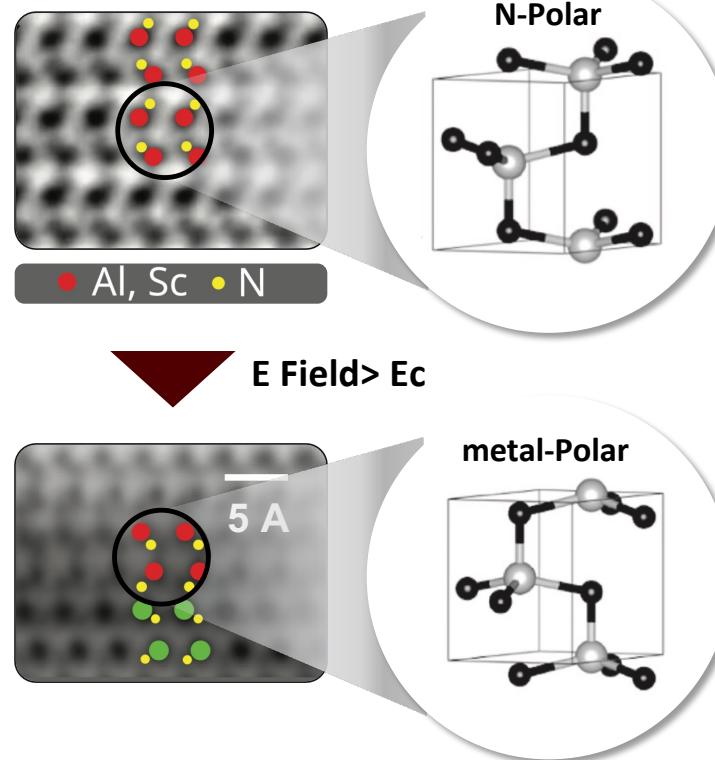
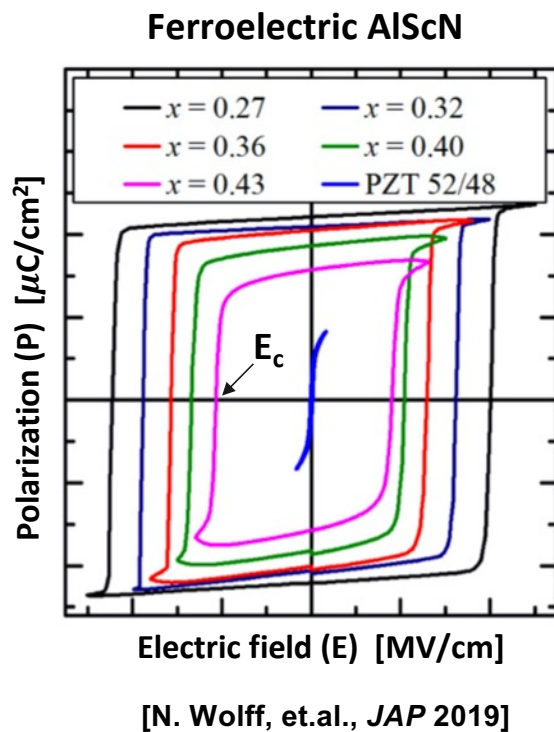
Measured Impedance Response



Summary of Contribution:

- ❑ Overcame a fundamental frequency limitation of AW technologies
- ❑ Invented a framework for multi frequency programmable AW devices: inhomogeneous piezoelectricity in multilayer ferroelectric heterostructures
- ❑ Enabled development of mm-Wave acoustic devices

Ferroelectric AlScN

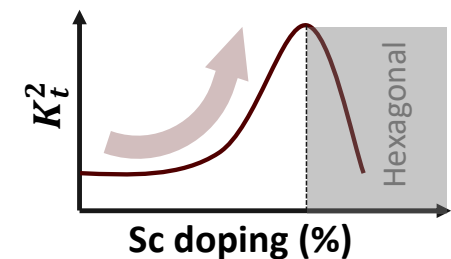


Process Development:

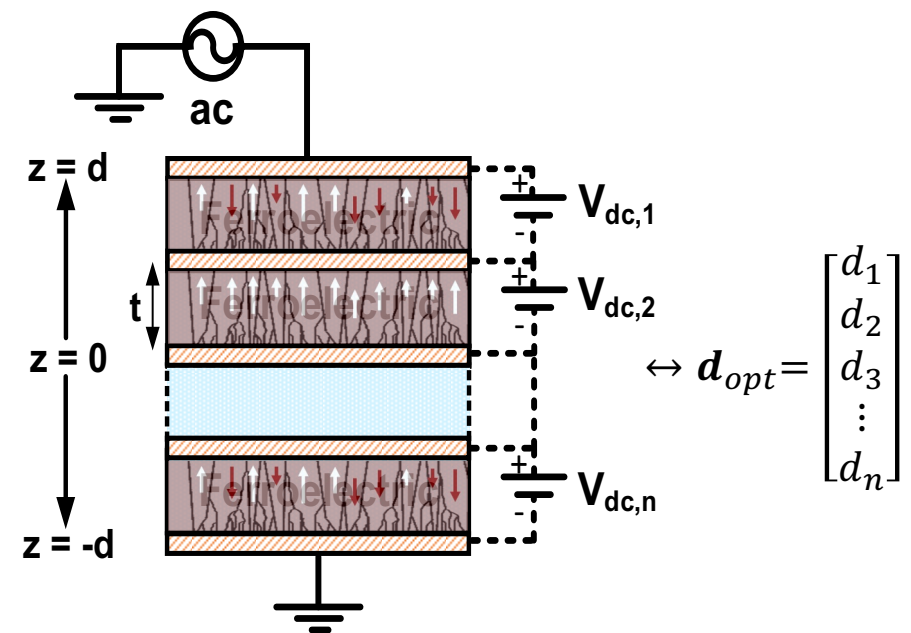
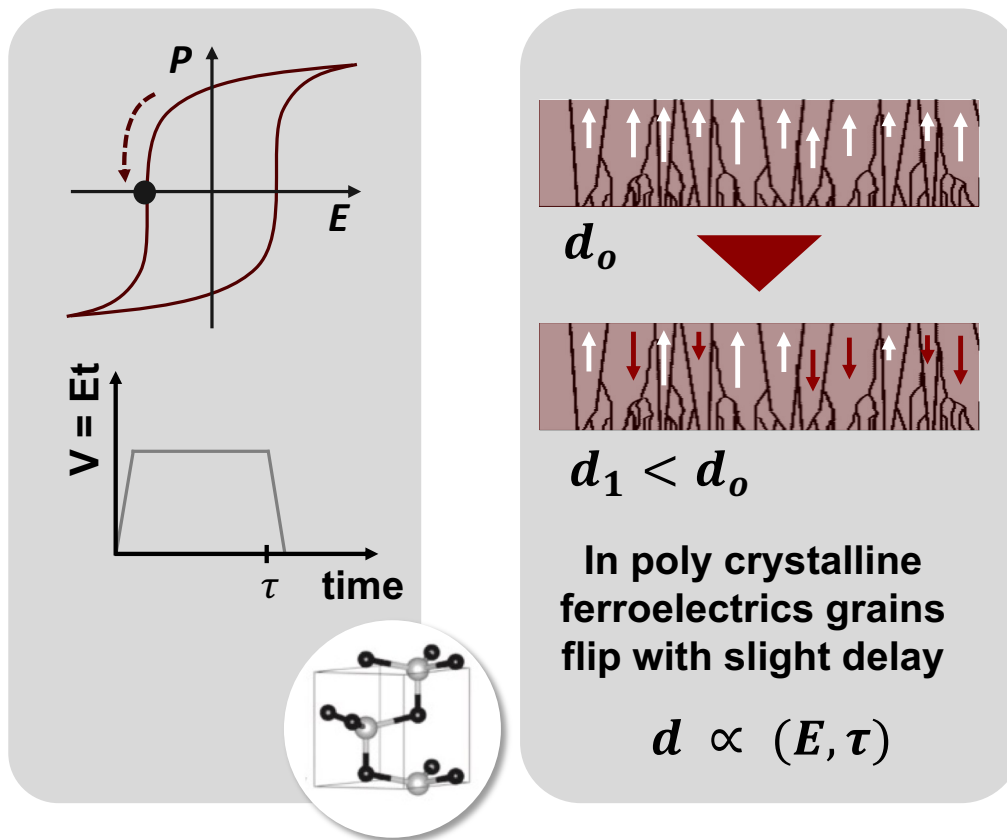
Ferroelectric AlScN thin film process is tailored for:

- low loss
- high electromechanical coupling
- low coercive field

Voltage dependent properties



Arbitrary Inhomogeneous Piezoelectric Medium: 2- In-plane grain polarization switching in ferroelectrics

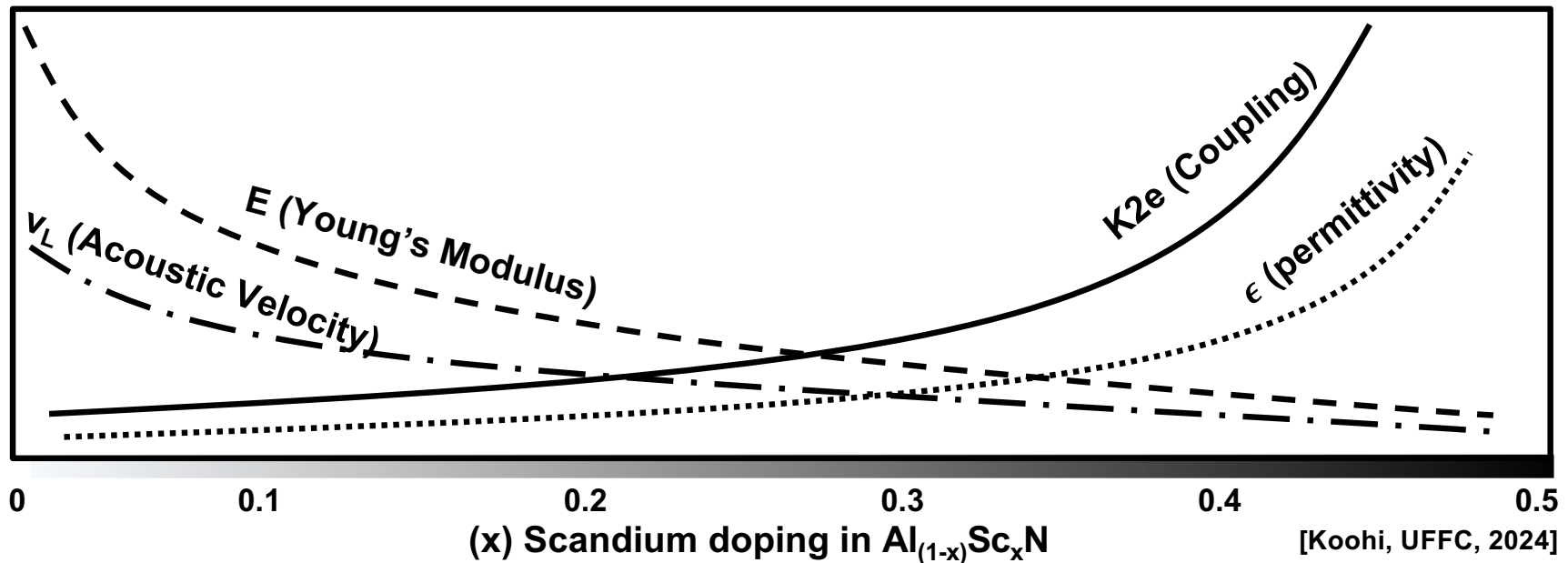
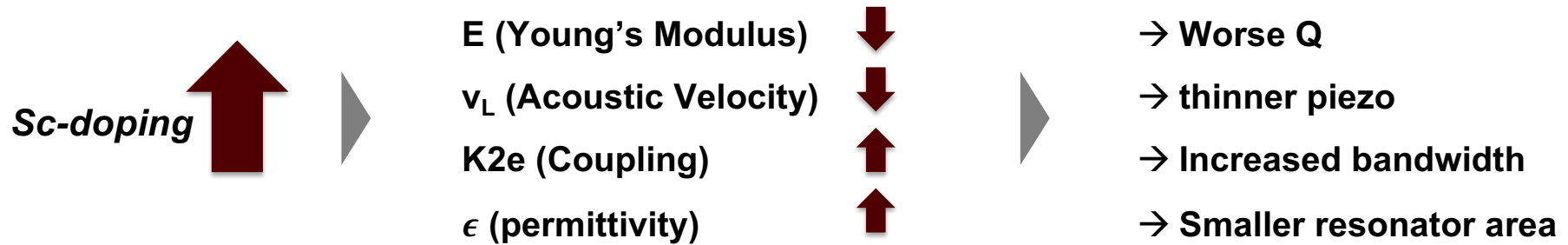


Bias to different voltage ($E \times \tau$) vectors or time vectors to create vertical inhomogeneous piezoelectricity

Material Properties of Sc-doped AlN?



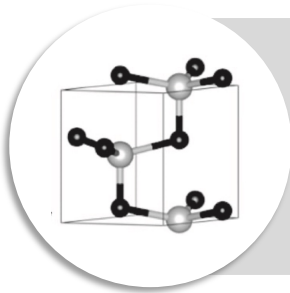
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mm-Wave Acoustic Integrated Circuits (AWIC)



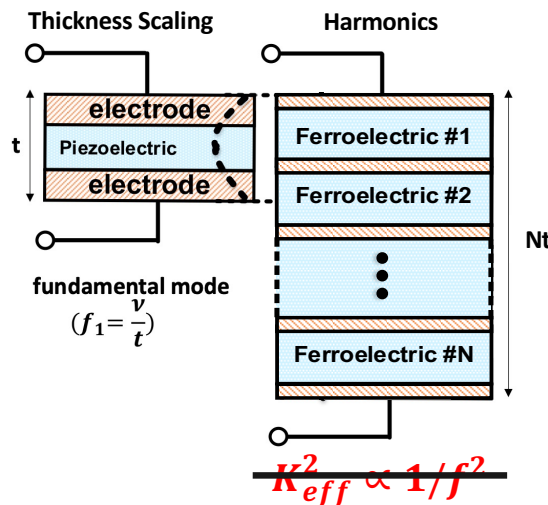
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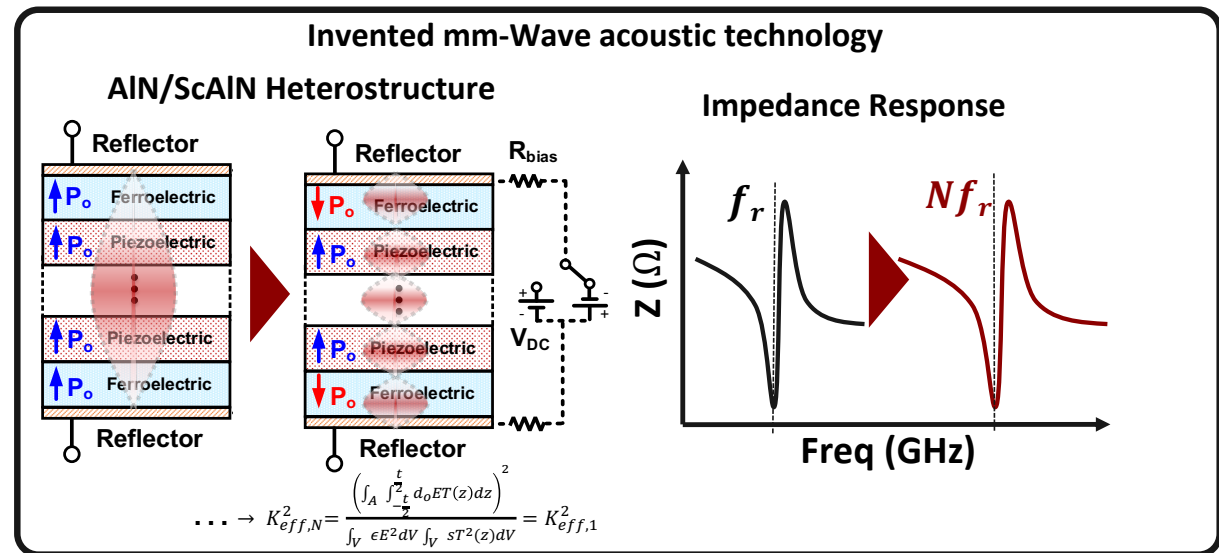
mm-Wave AW Technology:

- ❑ Invented mm-Wave acoustic resonator based on ferroelectric/piezoelectric heterostructures
- ❑ Experimentally demonstrated the first 20 GHz AW device with the highest FOM ($Q \times K^2 \times \text{freq}$ = 130×10^{10}) reported to date

Traditional mm-Wave solutions



Invented mm-Wave acoustic technology



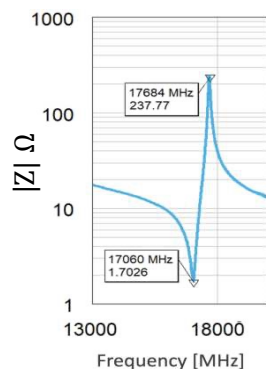
*Fabricated device image and measurement data are not shown due to Qorvo policy

[M. Koohi, et. al, US patent]

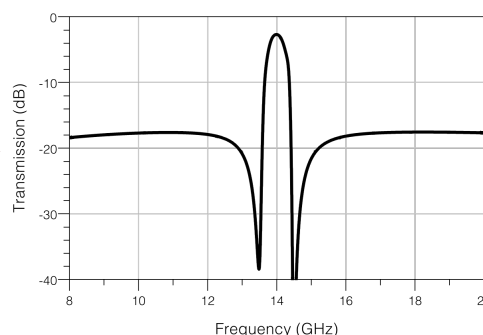
mm-Wave Acoustic Integrated Circuits (AWIC)

The **first bulk acoustic wave resonator** measured **above 10 GHz** and highest performance to date

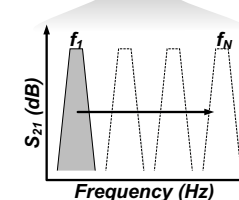
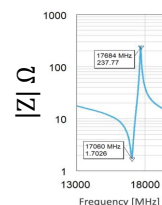
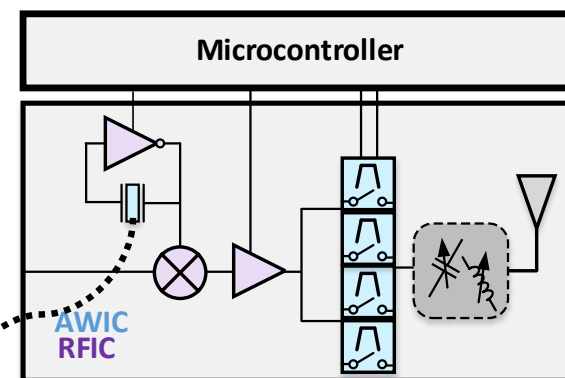
Parameter	Measured
f_s [GHz]	17
K_e^2 [%]	8.6
Q_s and Q_p	>10x LC



mm-Wave bandpass filters for 5G+



mm-Wave AWIC and RFIC heterogenous integration for high performance communications:

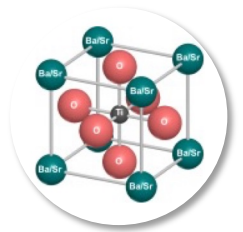


Conclusion



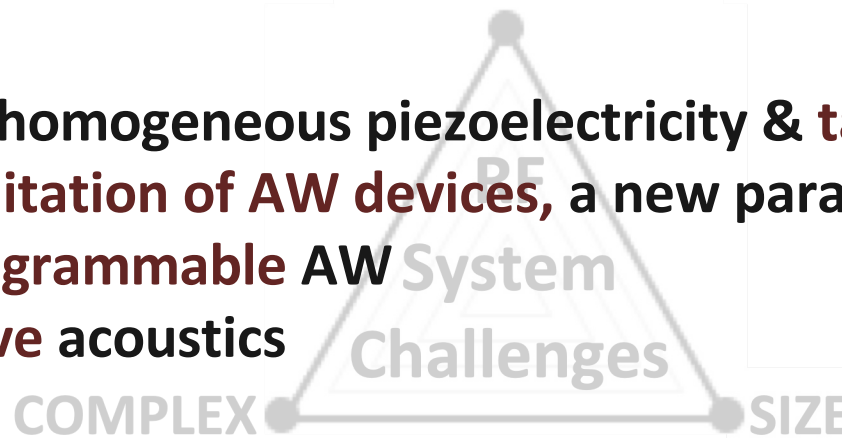
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❑ Exploited **electrostriction phenomenon** & developed a platform for **reconfigurable RF AW technologies**

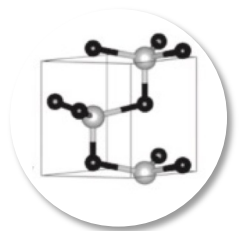


❑ Introduced inhomogeneous piezoelectricity & **tackled a fundamental frequency limitation of AW devices**, a new paradigm for:

- **Fully programmable AW System**
- **mm-Wave** acoustics



❑ Demonstrated inhomogeneous piezoelectricity **realized by ferroelectric domain switching** (e.g., in Al_{0.5}N_{0.5}) as a framework for mm-Wave acoustics required for 5G+ and 6G applications





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Q&A